Ozark Precision Manufacturing Network Proposal

Prepared by Cody Phillips, Network Designer, for Ozark Precision Manufacturing, a fictional company created for this portfolio project. This revised proposal aligns the headquarters and two branches around one user internet exit at headquarters, a private WAN to Branch 1, and an encrypted full tunnel from Branch 2. It preserves the original fifteen milestones and distinguishes the portfolio lab demonstration from deployment acceptance.

Contact: chphillips40@gmail.com | 501-316-9283

Project status: Proposed design with a planned Cisco Packet Tracer implementation. Headquarters represents the main business office, Branch 1 the production facility and Branch 2 the distribution office. This portfolio covers their business IT networks; factory control systems and PLC networks are outside this design. Configuration, screenshots and test results will be added after the lab is built. No production deployment or completed test results are claimed.

Project requirements

Requirement	Headquarters	Branch 1	Branch 2
LAN users	35	15	5
Phones	40	17	2
Servers	2	0	0
Corporate wireless	Yes	Yes	No
Guest wireless	Yes	No	No
Internet circuits	2	0	1
Private WAN circuits	1	1	0
PSTN service	1	1	0
VPN endpoint	HQ WAN router	None	Branch 2 router

HQ internet circuit 1 carries ordinary internet traffic and NAT. HQ circuit 2 carries the Branch 2 VPN underlay only. Branch 2 has an internet circuit for its VPN underlay; its user internet traffic is encrypted to HQ and exits through HQ circuit 1. These circuits do not constitute automatic internet failover.

HQ core redundancy covers first hop routing and dual switch paths. Single routers, access switches, server attachments and provider circuits remain possible failure points. No end to end high availability is promised.

Scope and equipment

The scope covers LAN routing and switching, WAN routing, security, VoIP and wireless at all three sites. Provider installation, building cabling, racks, power installation and UPS installation remain customer responsibilities.

Equipment	Quantity and allocation	Clarification
Cisco WS-C3560-24PS	2 at HQ	Layer 3 core; HSRP and PVST; intercore EtherChannel
Cisco WS-C2960-24TT	3 HQ; 1 B1; 1 B2	5 total; dual HQ uplinks; branch router trunks
Cisco CP-7960	40 HQ; 17 B1; 2 B2	59 full deployment; 7 representative lab phones
AccessPoint-PT	2 HQ; 1 B1	Simulation object; physical AP selection requires separate validation
Cisco 2901	HQ-INET-RTR	ipbasek9 and securityk9
Cisco 2911	HQ-WAN-RTR	ipbasek9 and securityk9
Cisco 2911	HQ-VOICE-RTR	ipbasek9 and uck9
Cisco 2911	B1-RTR	ipbasek9, securityk9 and uck9 as listed in original equipment inventory
Cisco 1941	B2-RTR	ipbasek9 and securityk9
Servers and endpoints	2 HQ servers; hosts and tablets	HQ Server 1 DHCP; Server 2 purpose requires confirmation

Capacity and power

The 2960-24TT access switches do not provide phone PoE. Supply compatible external phone power adapters in the physical bill of materials; in Packet Tracer attach each phone power adapter. Budget AP power separately. Port and power capacity must be verified against the actual equipment before purchase.

B1 needs 17 phone ports with up to 15 PCs behind phone PC ports, plus 1 AP port and an uplink. If PCs require separate switch ports, a 24 port switch is insufficient for 17 phones plus 15 PCs. HQ has 72 access ports for 40 phones with 35 PCs sharing phone PC ports; reserve guest AP and wired guest ports. B2 needs 2 phone ports and 5 PC ports, or shared phone PC ports where appropriate.

Validate CME platform and license capacity for 42 registered phones at HQ and 17 at B1; allow suitable directory number capacity. A seven phone simulation cannot prove full deployment capacity. Obtain exact firmware, image, licensing and supported command information before physical deployment.

Addressing and default gateways

All LAN networks use /24 masks (255.255.255.0). Preserve the source VLAN IDs and site networks. Use .1 and .2 as HQ core SVI addresses and .100 as the HSRP gateway in every HQ VLAN. Branch router gateways are .1 in each branch VLAN.

Site and VLAN	Subnet	Default gateway
HQ 100 Management	192.168.110.0/24	192.168.110.100
HQ 192 Data	192.168.10.0/24	192.168.10.100
HQ 10 Voice	10.10.10.0/24	10.10.10.100
HQ 172 Guest	172.16.10.0/24	172.16.10.100
B1 100 Management	192.168.120.0/24	192.168.120.1
B1 192 Data	192.168.20.0/24	192.168.20.1
B1 10 Voice	10.10.20.0/24	10.10.20.1
B2 100 Management	192.168.130.0/24	192.168.130.1
B2 192 Data	192.168.30.0/24	192.168.30.1
B2 10 Voice	10.10.30.0/24	10.10.30.1

HQ-INET-RTR uses 192.168.110.253 for management and 192.168.10.253 for data/transit. HQ-WAN-RTR uses 192.168.110.254 for management and 192.168.10.254 for data/transit. HQ-VOICE-RTR uses 192.168.110.252 for management and 10.10.10.10 for voice. Existing topology router labels are management addresses, not interchangeable data next hops.

HQ access switch management addresses remain 192.168.110.241, .242 and .243. B1-SW1 is 192.168.120.254; B2-SW1 is 192.168.130.254. Server 1 is 192.168.10.50; reserve .51 for Server 2. AP addresses are proposed as HQ corporate 192.168.10.60, HQ guest 172.16.10.60, and B1 corporate 192.168.20.60; verify management support on the chosen AP object.

Reserve .1 through .100 and .241 through .254 for infrastructure. Dynamic ranges are .101 through .200, with 100 leases per scope. HQ data and guest scopes run on Server 1; HQ voice runs on the HQ voice router. Branch data and voice scopes run on their local routers. Management addressing is static.

Physical interfaces and switching

Device	Interface	Role and VLANs
HQ-INET-RTR	G0/0 trunk	G0/0.100 management; G0/0.192 data/transit, NAT inside
HQ-INET-RTR	G0/1	Internet circuit 1, NAT outside
HQ-WAN-RTR	G0/0 trunk	G0/0.100 management; G0/0.192 data/transit
HQ-WAN-RTR	G0/1	Private WAN to provider
HQ-WAN-RTR	G0/2	Internet circuit 2; VPN crypto map
HQ-VOICE-RTR	G0/0 trunk	G0/0.100 management; G0/0.10 voice
HQ-VOICE-RTR	G0/1	PSTN provider IP handoff
B1-RTR	G0/0 trunk	Subinterfaces .100, .192 and .10
B1-RTR	G0/1	Private WAN to provider
B1-RTR	G0/2	PSTN provider IP handoff
B2-RTR	G0/0 trunk	Subinterfaces .100, .192 and .10
B2-RTR	G0/1	Internet underlay and VPN crypto map

All LAN subinterfaces use dot1Q encapsulation for their VLAN ID. No IP is assigned to the parent LAN interface. Branch switches trunk to their own local router. Install one access switch at each branch, correcting the original references to three switches and to Branch 1 from the Branch 2 milestone.

Use a two member LACP intercore EtherChannel. Each HQ access switch has one independent trunk to each core. Do not create an EtherChannel across two independent 3560 cores. Allow VLANs 10, 100, 172 and 192 on intercore and HQ access trunks. Router trunks allow only their required VLANs. Use a consistent unused native VLAN such as 999; assign it no SVI or hosts.

Match HSRP active roles with spanning tree root roles: Core 1 primary for VLANs 10 and 192; Core 2 primary for VLANs 100 and 172. Use root priority 24576 on the primary and 28672 on the secondary, with HSRP priorities 110 and 100 plus preempt. Track the relevant core uplink where supported; test failover rather than assuming it.

Retain VTP as a lab objective: HQ Core 1 is the designated VLAN administrator in server mode; Core 2 may remain server mode but must not make independent changes. HQ access switches are clients in the same domain and password. Verify VTP version and reset revision on new devices before attaching trunks. Branch switches are transparent and have VLANs created locally.

Routing and BGP

Router or switch	Destination	Next hop
Both HQ cores	Default	192.168.10.253
Both HQ cores	All three B1 and all three B2 /24 networks	192.168.10.254
HQ Internet router	B1 and B2 /24 networks	192.168.10.254
HQ Internet router	HQ voice and HQ guest	192.168.10.100
HQ Internet router	Default	Circuit 1 ISP next hop
HQ WAN router	HQ voice and HQ guest	192.168.10.100
HQ WAN router	B1 networks	BGP via private WAN
HQ WAN router	B2 networks and B2 public peer /32	Circuit 2 ISP next hop via G0/2
HQ WAN router	Default	192.168.10.253
HQ voice router	Default	10.10.10.100
B1 router	Default	Private WAN provider next hop
B2 router	Default and HQ public peer /32	B2 ISP next hop via G0/1

Every static next hop must be on a connected subnet or have a verified recursive route that selects the intended interface. Use the local ISP gateway for underlay routes. Do not use a remote public peer as though it were a locally attached gateway. Keep HQ data and management as connected networks on the HQ Internet and WAN routers.

BGP ownership and loop prevention

Preserve the specified customer ASN 65123 at HQ and B1. The provider must supply its ASN, neighbors, handoff masks, routing policy and confirmation of same customer ASN support. A provider path containing 65123 would normally be rejected at the other customer site. Prefer provider AS override with appropriate loop prevention. If provider AS override is unavailable, a lab alternative is allowas-in 1 on the CE provider neighbor, with strict received prefix filters and no uncontrolled redistribution. Confirm command support before use. If neither is available, an documented distinct B1 ASN is required.

Advertise HQ management, data and voice plus all three B2 networks from the HQ WAN router, where exact routes exist in its routing table. Advertise only the three B1 networks from B1. Do not advertise guest or a default route into BGP unless the provider design explicitly requires it. B1 uses its static provider default for internet via HQ; the provider must forward that default traffic toward HQ. Apply prefix filters so route acceptance does not weaken loop controls.

VPN NAT and firewall behavior

The VPN is full tunnel for Branch 2. Traffic from each B2 private subnet to any nonlocal destination must enter the tunnel, including public DNS, internet web destinations, B1 and HQ. A VPN covering only HQ and B2 private subnets would break the single user internet exit requirement.

Mirrored encryption selectors

```
! HQ WAN router crypto ACL
ip access-list extended B2-VPN
 permit ip any 192.168.130.0 0.0.0.255
 permit ip any 192.168.30.0 0.0.0.255
 permit ip any 10.10.30.0 0.0.0.255
! Branch 2 crypto ACL
ip access-list extended HQ-VPN
 permit ip 192.168.130.0 0.0.0.255 any
 permit ip 192.168.30.0 0.0.0.255 any
 permit ip 10.10.30.0 0.0.0.255 any
```

A crypto ACL selects encryption; it does not authorize corporate access. Apply separate access policies to decrypted traffic. Confirm that Packet Tracer supports these selectors and full tunnel forwarding; otherwise document the limitation and document an equivalent lab demonstration. Do not silently enable local internet breakout.

HQ WAN router sends B2 private destinations toward G0/2, where the crypto map encrypts them. Its default route sends decrypted B2 internet traffic to HQ-INET-RTR over the HQ data VLAN. HQ-INET-RTR overloads HQ, B1 and B2 data sources plus HQ guest onto circuit 1. Include other private sources only when their access is authorized. Return routes for B2 point to the HQ WAN router. Do not configure user internet NAT on B2 or HQ circuit 2.

VPN outer traffic must be routable between public peers through their local ISP gateways. Match authentication, shared key, encryption, integrity, DH group, transform set and lifetimes at both ends. Use a unique secret supplied privately, never a published sample password. Use supported IKE/IPsec algorithms in the lab; older ISAKMP commands are a simulator exercise and require a separate current platform security review for production.

At HQ circuit 2 accept VPN transport from B2; at B2 accept VPN transport from HQ, correcting the original instruction to accept traffic from B2 itself. Account for UDP 500, UDP 4500 when NAT traversal is used, and ESP. Check whether the platform interface ACL also processes decrypted inner packets. Do not apply an outer-only ACL that drops valid decrypted traffic, or permit spoofed plaintext solely because it uses a B2 source address. Use an IPsec-aware policy where supported and verify both allowed and denied cases.

HQ circuit 1 uses stateful inspection and a compatible outside ACL for return traffic, DNS, web and permitted diagnostic ICMP. Disable CDP on external interfaces. Deny unsolicited inbound access and router administration from the internet. Restrict SSH to a designated management host and use local secrets, time synchronization and saved configurations.

DHCP wireless and voice

Scope	Gateway	Special settings
HQ data 192.168.10.0/24	192.168.10.100	Server 1; .101-.200; DNS 8.8.8.8
HQ guest 172.16.10.0/24	172.16.10.100	Server 1; .101-.200; DNS 8.8.8.8
HQ voice 10.10.10.0/24	10.10.10.100	HQ voice router; option 150 = 10.10.10.10
B1 data 192.168.20.0/24	192.168.20.1	B1 router; .101-.200; DNS 8.8.8.8
B1 voice 10.10.20.0/24	10.10.20.1	B1 router; option 150 = 10.10.20.1
B2 data 192.168.30.0/24	192.168.30.1	B2 router; .101-.200; DNS 8.8.8.8
B2 voice 10.10.30.0/24	10.10.30.1	B2 router; option 150 = 10.10.10.10

Configure ip helper-address 192.168.10.50 on both HQ guest SVIs. Place DHCP permits ahead of guest denial rules; verify initial lease broadcasts, relayed replies and renewals. Do not give guests unrestricted access to Server 1. Disable unneeded helper forwarding protocols where supported.

Guest policy allows necessary DHCP and internet services and denies corporate and infrastructure networks, including all 10.0.0.0/8, 172.16.0.0/12 and 192.168.0.0/16 destinations after the narrow DHCP exception. Apply inbound on both guest SVIs. Restrict access to infrastructure public addresses as well. Test isolation while either core is active. Internet-only routed ACLs do not guarantee wireless client isolation within a shared guest VLAN.

Corporate HQ AP attaches to Core 2 in VLAN 192; guest AP attaches to HQ Access 3 in VLAN 172; B1 AP attaches to B1-SW1 in VLAN 192. Use separate corporate and GUEST SSIDs, WPA2-PSK/AES as the lab minimum, unique strong passphrases and nonoverlapping channels as appropriate. Test one corporate and one guest HQ tablet and two B1 tablets. B2 has no AP.

Phone ports use access VLAN 192 and voice VLAN 10. PCs may connect behind phones. Enable PortFast and BPDU Guard only on endpoint ports, not trunks. HQ CME serves HQ and B2; B1 CME serves B1. HQ lab extensions are 1001-1003, B1 2001-2002 and B2 3001-3002. Full deployment ranges are proposed as 1001-1040, 2001-2017 and 3001-3002.

HQ dial peer 2... routes to 10.10.20.1. B1 requires separate 1... and 3... dial peers to 10.10.10.10, including the omitted B2 range. B2 registers with HQ and uses the HQ PSTN service. Classify voice signaling and RTP on both private WAN ends; shape to the agreed service rate before applying a bounded priority queue. Confirm codec, concurrent calls and bandwidth before selecting queue rates. Public internet QoS is not guaranteed.

Milestones I through V

I Project kickoff

Confirm portfolio lab versus hypothetical deployment acceptance, topology, equipment and power, provider dependencies, addresses, simulator version, schedule and contact details. Agree maintenance windows, rollback triggers and evidence requirements. Define the simulated equipment, cabling and provider handoffs before configuration. Record the chosen addressing plan in the lab notes.

II HQ core switching

Install two cores; configure hostnames, management, SSH and time; enable IP routing; create VLANs and all SVIs. Configure LACP intercore trunks, designated VTP administration, HSRP for all four routed VLANs and aligned PVST roots. Configure access and trunk ports, default route to HQ-INET data .253 and branch routes to HQ-WAN data .254. Validate independent HSRP and spanning tree failover.

III HQ access switching

Install three switches; set management VLAN 100 at .241-.243 and gateway .100; configure VTP clients and dual trunks, VLAN pruning and endpoint ports. Configure data and voice membership, guest ports, phone power and edge safeguards. Verify VLAN propagation and forwarding before connecting all endpoints.

IV HQ Internet router

Install the 2901 and validate security licensing. Configure G0/0.100 and .192, with NAT inside only on data/transit; G0/1 is circuit 1 NAT outside. Implement NAT overload, stateful inspection and outside ACL. Add explicit return routes for branch subnets through HQ-WAN and for voice and guest through the HQ data HSRP gateway. Use the ISP gateway as default. Verify internet traffic and reject unsolicited inbound connections.

V HQ WAN router

Install the 2911 and validate security licensing. Configure LAN subinterfaces, G0/1 private WAN and G0/2 VPN internet underlay. Establish filtered BGP in ASN 65123 with agreed same-AS handling. Configure voice QoS for private WAN. Implement matched IPsec policies, full tunnel selectors and VPN-only outside policy. Set default toward HQ-INET data .253, HQ voice/guest routes toward data .100, B2 subnet routes and peer /32 toward circuit 2 ISP gateway. Validate bidirectional B1 and B2 forwarding.

Milestones VI through XI

VI HQ voice router

Install the 2911 with UC licensing; configure management and voice LAN subinterfaces, default to voice HSRP .100, HQ voice DHCP and option 150. Configure CME for the agreed endpoint count and extensions. Use G0/1 for provider voice service and apply provider-specific ACL. Configure B1 and supported outbound dial peers for 7 digit local, 10 digit long distance and toll free, international, 411 and 911 patterns. Document provider requirements and use simulated test numbers only. Inbound PSTN translation, SIP, DSP and other unsupported simulator tasks require a design record and are not claimed as lab-tested.

VII HQ phones and hosts

For the representative lab, power and register three HQ phones, one per access switch, and connect three PCs behind them. Record MAC bindings and ephones. Configure Server 1 data DHCP, gateway, DNS and lease range. Test internal calls, simulated external test number 8885551111, local connectivity, public test server reachability, DNS and HTTP. Physical deployment acceptance additionally requires all 40 phones and 35 users or documented exhaustive evidence.

VIII HQ wireless

Install corporate and guest APs on their assigned access VLANs. Configure separate SSIDs and WPA2/AES settings. Add HQ guest DHCP scope and helpers on both cores; enforce guest isolation before joining tablets. Test corporate access and guest internet, plus failed guest access to all sites and infrastructure, using each core as active gateway.

IX Branch 1 route and switch

Install one 2911 and one 2960. Configure three router LAN subinterfaces, local DHCP, CME and extensions, management gateway and transparent VTP. G0/1 is private WAN; G0/2 is PSTN. Establish filtered BGP in ASN 65123 with the agreed same-AS remedy. Route default to provider, and configure private WAN QoS. Configure both HQ and B2 voice dial peers and supported outbound PSTN peers for local, long distance, toll free, international, 411 and 911 patterns. Translation, inbound SIP and DSP tasks remain design-only where unsupported.

X Branch 1 phones and hosts

Power and register two representative B1 phones; connect two PCs behind them. Test calls to HQ and B2, simulated external dialing, B1/HQ/B2 data paths, internet via HQ and DNS/HTTP. Deployment acceptance requires 17 phones and 15 users. Do not substitute two working phones for the contractual count.

XI Branch 1 wireless

Install one corporate AP on B1 data VLAN 192; configure SSID, passphrase, AES and channel. Join two tablets and verify corporate and internet access via HQ. Use simulated AP and tablet objects and record their test results.

Milestones XII through XV

XII Branch 2 route and switch

Install one 1941 and one 2960. Configure local management, data and voice VLANs on G0/0 subinterfaces, transparent VTP, switch management .254 and correct local router trunk. Configure data/voice DHCP, with option 150 pointing to HQ CME. Configure G0/1 underlay and peer policy accepting HQ VPN transport. Default and HQ peer /32 use the local ISP gateway. Match full tunnel selectors and crypto map; do not NAT locally. Verify encrypted traffic to HQ, B1 and internet.

XIII Branch 2 phones and hosts

Power two B2 phones and register them on HQ CME as 3001 and 3002. Test B2, HQ and B1 calls and simulated HQ PSTN breakout. Connect two representative B2 PCs and verify their local subnet, all required remote subnets, DNS and HTTP through HQ NAT. For deployment acceptance, validate all five B2 users. Correct the original instruction to test around the Branch 1 network as though it were local.

XIV Testing support and confirmation

Complete the acceptance matrix on the following page and attach evidence to each result. Record pass, fail, blocked or design-only status; unsupported simulator features are never marked pass. Investigate failures, update configurations and rerun affected tests. Reserve a two-day troubleshooting and documentation review period after the initial lab build.

XV Project closure

Review the portfolio deliverables against the fictional company requirements. Hand over the Packet Tracer file when available, saved device configurations, diagram, IP and VLAN plan, provider handoff record, test results, backups and remaining limitations. Mark portfolio completion only when supported tests pass and limitations are documented. Update the scope record if the lab design changes.

Implementation order and rollback

Build switching and addressing first, then local DHCP and routing, then private WAN/BGP, VPN/full tunnel and NAT, then voice and wireless. Save known working configurations before each stage. If a stage breaks a previously accepted path, restore its prior configuration, preserve evidence and correct the change before continuing. Simulator-only ISP, DNS, web and PSTN devices must be supplied by the portfolio lab or built as approved test fixtures.

Acceptance matrix and evidence

Test	Required evidence	Pass condition
VLANs and trunks	show vlan brief; show interfaces trunk; port map	Correct memberships; allowed/native VLANs agree
EtherChannel	show etherchannel summary	Both intercore members bundled; no cross-core access bundle
HSRP and PVST	show standby brief; show spanning-tree; failover log	Roles align; traffic resumes after core/uplink loss within agreed target
Routing and BGP	show ip route; show ip bgp summary; route tables at each site	All required prefixes accepted; return paths exist; no routing loop
DHCP	Leases and endpoint settings in every active scope	Correct gateway, DNS, option 150; no reserved address leased
VPN corporate paths	show crypto isakmp sa; show crypto ipsec sa; ping tests	HQ/B1/B2 paths work; encrypt and decrypt counters increase
B2 internet full tunnel	B2 web/DNS test; crypto counter deltas; HQ NAT table	Internet packets are encrypted to HQ and NAT at circuit 1
Internet NAT and firewall	show ip nat translations/statistics; allowed and denied tests	Authorized users online; unsolicited inbound blocked
Guest isolation	Guest tests to every corporate subnet and infrastructure address	DHCP/DNS/web succeed; corporate access fails under both cores
Voice	Registration, extension call matrix and simulated PSTN call record	HQ/B1/B2 two-way calls and audible media in supported lab
Wireless	SSID/security evidence and tablet addressing	Corporate access works; guest restriction holds
Capacity and endpoint count	Port inventory, phone power and CME capacity checks	59 phones/55 LAN users for deployment; 7 lab phones only by explicit acceptance
Unsupported functions	Feature support record and project reviewer exceptions	Design-only items labeled and accepted; no fabricated test results
Handover	Saved configurations and deliverable checklist	Files open; configurations saved; exceptions recorded

Set measurable lab failover, delay, loss and call-quality targets before testing; no numerical service level is invented in this proposal. If the simulated topology uses a simulated Google DNS/web server, use its configured addresses and name records. A Packet Tracer browser test does not prove live internet access.

Portfolio scope responsibilities and assumptions

Project ownership and audience

Cody Phillips is the Network Designer for this proposed portfolio project. Ozark Precision Manufacturing is fictional, and the intended audience is potential employers reviewing network design, implementation planning and troubleshooting skills. Cisco Packet Tracer is the planned lab platform. The technical baseline is adapted from the supplied Full Stack Networking Project documents.

Business scenario

Headquarters is the main business office with 35 LAN users and 40 phones. Branch 1 is the production facility business office with 15 LAN users and 17 phones. Branch 2 is the distribution office with 5 LAN users and 2 phones. These requirements describe the scenario, not actual employment, a real customer or measured equipment in service.

Implementation responsibilities

Cody Phillips will build the lab, record actual device and simulator versions, configure the supported features, troubleshoot the traffic paths and collect test evidence. Proposed provider services are represented by simulated routers and test servers. Project contact: chphillips40@gmail.com; 501-316-9283. No actual procurement, provider service order, customer signature or payment is required for the portfolio lab.

Assumptions and exclusions

The proposal assumes provider handoffs, racks, installed cabling, phone power and sufficient device capacity in a hypothetical physical deployment. Building cabling, power installation, physical procurement, production deployment and factory control networks are outside this portfolio implementation. Unsupported simulator functions remain design-only and are documented explicitly.

Planning and review

Use the fifteen technical milestones to build the lab and the acceptance matrix to record outcomes. Track dates and effort as the work is performed; no estimated fee, commercial billing term or binding contract is included. Revise the network plan when test evidence exposes a limitation, recording the reason and the affected requirements.

Portfolio completion

The project is complete when the planned Packet Tracer file opens, supported functions pass documented tests, remaining limitations are explained and the diagram matches the saved configurations. Seven representative phones are the portfolio demonstration target. The original 59-phone business requirement remains a capacity-planning scenario and is not claimed as deployed or fully validated.

Evidence to add after implementation: final Packet Tracer file, saved configurations, screenshots, routing and security test results, call tests, failure recovery observations and a short explanation of design choices.

Completion form and open decisions

Deliverable	Evidence reference	Status and date
HQ redundant core switching		
Single user internet exit through HQ circuit 1		
HQ corporate LAN and wireless for 35 users		
HQ wired and wireless guest internet isolation		
HQ and B1 private WAN		
B1 corporate LAN and wireless for 15 users		
HQ and B2 full tunnel VPN		
B2 corporate LAN for 5 users		
HQ and B1 supported PSTN demonstration		
59 deployment phones or explicitly accepted 7 phone lab		
Testing evidence and agreed exceptions		
Lab handover and documentation review		

Decisions that require supplied facts

Before implementation, define simulated public IPs, masks and gateways for three internet handoffs; private WAN handoff IPs, provider ASN and forwarding policy; same-AS BGP support; PSTN peer IPs, masks, transport, DIDs and codec; Packet Tracer version and starter file; administrator workstation address; modeled WAN rates and call concurrency; Server 2 role; DNS/web test fixture; failover target; lab versions, build dates and evidence references. These are dependencies, not guessed provider facts.

The revised diagram includes clearly labeled documentation-only ISP and WAN addresses for a self-contained example. Use actual starter-file addresses when provided. Do not mix the example underlay with existing provider devices. Do not place real emergency calls to test a portfolio lab; use only simulated or provider-authorized test procedures.

Portfolio author: Cody Phillips Project role: Network Designer

Implementation dates _____ Limitations and evidence references _____

Correction register and technical references

Source location	Correction
Summary and provider list	HQ has two internet circuits; one user exit and one VPN transport circuit
Core static routes	Both B1 and B2 included; data next hops explicitly distinguished from management
HQ WAN static routes	Voice/guest route through adjacent data VIP; B2 underlay through local ISP gateway
B2 VPN and routes	Full tunnel selectors include public destinations; default uses local ISP gateway
B1 router tasks	Private WAN G0/1; PSTN G0/2; added dial peer for B2 extensions
BGP at HQ and B1	Same ASN retained with explicit AS override or filtered allowas-in dependency
B1 and B2 switch tasks	One switch at each branch; B2 trunks to B2 router
B2 outside filter	Allow HQ peer transport instead of self-peer instruction
Endpoint and completion counts	Seven lab phones separated from 59 deployment phones; 55 LAN users retained
DHCP, guest and NAT	Ranges/exclusions/relay, bidirectional return routes and isolation made explicit
Equipment inventory	Correct ipbasek9 spelling; phone power and CME capacity checks added
Topology and presentation	Added full interface/IP plan; single-exit flow and limits communicated consistently

Technical references checked 11 October 2026. The supplied three project documents are the requirements baseline. Cisco references support design checks; actual simulator command availability must still be verified.

Cisco Configure Allowas-in Feature in BGP: https://www.cisco.com/c/en/us/support/docs/ip/border-gateway-protocol-bgp/112236-allowas-in-bgp-config-example.html

Cisco Catalyst 2960 Switch Hardware Installation Guide Technical Specifications: https://www.cisco.com/c/en/us/td/docs/switches/lan/catalyst2960/hardware/installation/guide/2960_hg/higspec.html

Cisco Configure Policy Based Dynamic Site to Site VPN Tunnel: https://www.cisco.com/c/en/us/support/docs/security/dynamic-multipoint-vpn-dmvpn/226162-configure-policy-based-dynamic-site-to.html